

Name: SOLUTIONS

GCSE (1 - 9)

The Cosine Rule



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Solutions by

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These questions are compiled from *released* [Pearson Edexcel GCSE maths exam papers](#)

They contain exam questions from up until the **November 2021** exam series

More recent exam papers do exist but are withheld for mock exams

As new exam papers are released these worksheets will gradually be updated

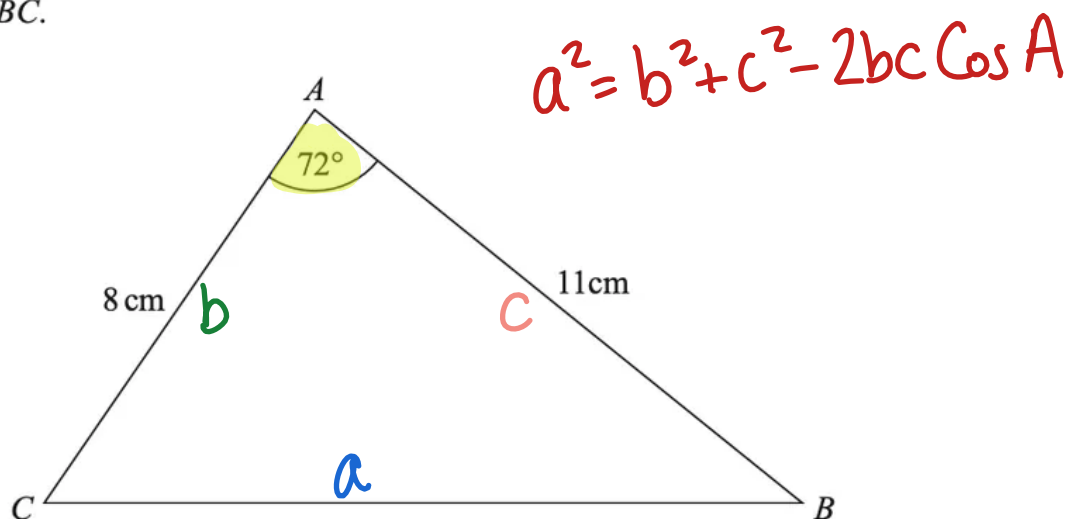
The **most current** version of these worksheets will always be available at: kennethstafford.co.uk

These worksheets were made using examq.co.uk

Question	Where the question appeared
1	Edexcel, GCSE, Autumn 2021, Paper 2H, Q15
2	Edexcel, GCSE, Autumn 2019, Paper 3H, Q18
3	Edexcel, GCSE, Summer 2019, Paper 3H, Q23
4	Edexcel, GCSE, Autumn 2018, Paper 3H, Q16
5	Edexcel, GCSE, Summer 2018, Paper 3H, Q17
6	Edexcel, GCSE, Autumn 2017, Paper 3H, Q17
7	Edexcel, GCSE, Summer 2017, Paper 1H, Q2

1

Here is triangle ABC .



(a) Find the length of BC .

Give your answer correct to 3 significant figures.

$$a^2 = 8^2 + 11^2 - 2 \times 8 \times 11 \times \cos 72^\circ$$

$$a^2 = 64 + 121 - 176 \cos 72$$

$$a^2 = 130.613$$

$$a = \sqrt{130.613} = 11.4286$$

$$\underline{\underline{11.4}} \text{ cm}$$

(3)

(b) Find the area of triangle ABC .

Give your answer correct to 3 significant figures.

$$\text{Area} = \frac{1}{2} \times b \times c \times \sin A$$

$$= \frac{1}{2} \times 8 \times 11 \times \sin 72^\circ$$

$$= 44 \times \sin 72 = \underline{\underline{41.8}} \text{ cm}^2$$

(2)

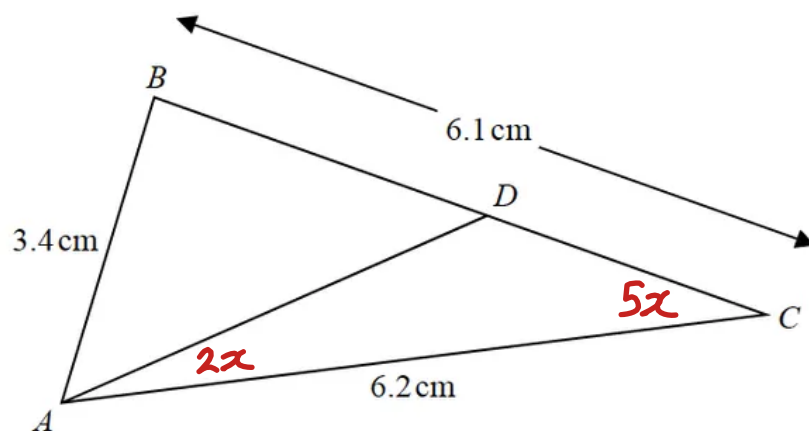
(Total for Question 15 is 5 marks)

[Edexcel, GCSE, Autumn 2021, Paper 2H, Q15]

The Cosine Rule (Questions from Past Papers)

2

The diagram shows triangle ABC .



$$AB = 3.4 \text{ cm} \quad AC = 6.2 \text{ cm} \quad BC = 6.1 \text{ cm}$$

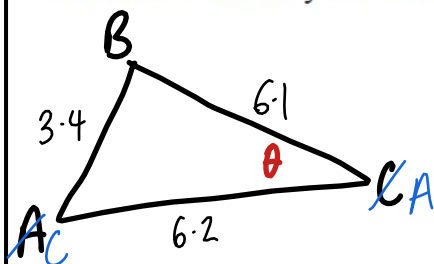
D is the point on BC such that

$$\text{size of angle } DAC = \frac{2}{5} \times \text{size of angle } BCA$$

Calculate the length DC .

Give your answer correct to 3 significant figures.

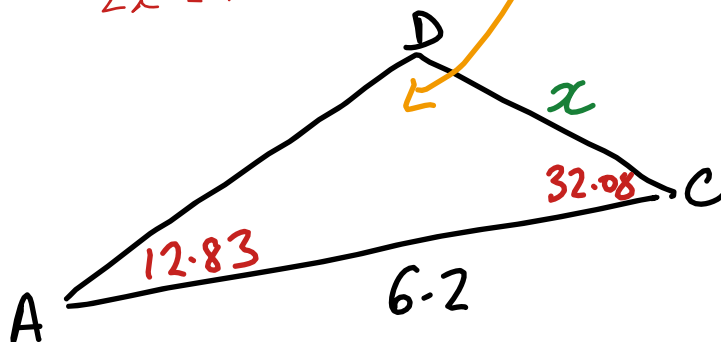
You must show all your working.



$$5x = 32.08$$

$$2x = 12.83^\circ$$

$$180 - 44.91 = 135.08$$



$$\frac{x}{\sin(12.83)} = \frac{6.2}{\sin(135.08)}$$

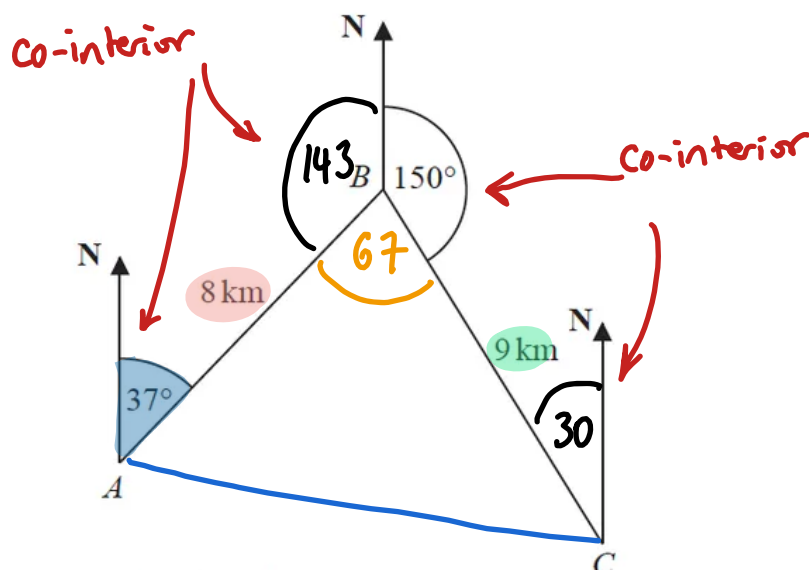
$$x = \frac{6.2 \sin(12.83)}{\sin(135.08)} = 1.95 \text{ cm}$$

(Total for Question 18 is 5 marks)

[Edexcel, GCSE, Autumn 2019, Paper 3H, Q18]

3

The diagram shows the positions of three towns, Acton (A), Barston (B) and Chorlton (C).



Barston is 8 km from Acton on a bearing of 037°

Chorlton is 9 km from Barston on a bearing of 150°

Find the bearing of Chorlton from Acton.

Give your answer correct to 1 decimal place.

You must show all your working.

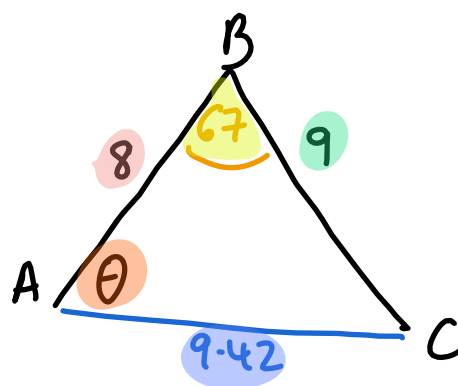
$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$a^2 = 9^2 + 8^2 - 2 \times 9 \times 8 \times \cos 67$$

$$a^2 = 88.7347$$

$$a = \sqrt{88.7347}$$

$$a = 9.42$$



$$\frac{\sin \theta}{9} = \frac{\sin 67}{9.42}$$

$$\sin \theta = \frac{9 \sin 67}{9.42}$$

$$\theta = \sin^{-1} \left(\frac{9 \sin 67}{9.42} \right)$$

$$\theta = 61.58$$

$$37 + 61.58 = 098.6^\circ$$

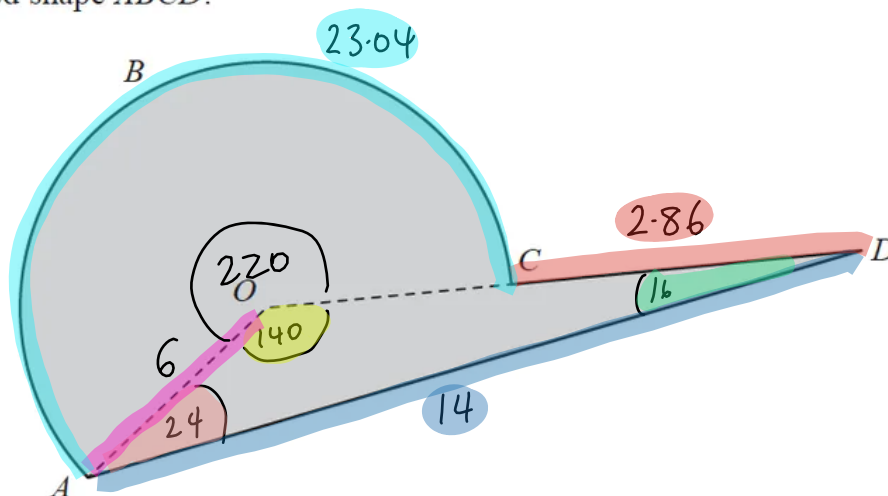
(Total for Question 23 is 5 marks)

[Edexcel, GCSE, Summer 2019, Paper 3H, Q23]

The Cosine Rule (Questions from Past Papers)

4

Here is a shaded shape $ABCD$.



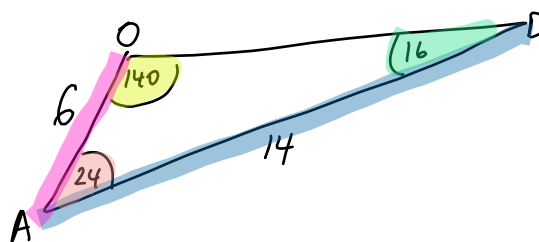
The shape is made from a triangle and a sector of a circle, centre O and radius 6 cm.
 OCD is a straight line.

$$AD = 14 \text{ cm}$$

$$\text{Angle } AOD = 140^\circ$$

$$\text{Angle } OAD = 24^\circ$$

Calculate the perimeter of the shape.
 Give your answer correct to 3 significant figures.



$$OD^2 = 6^2 + 14^2 - 2 \times 6 \times 14 \times \cos(24)$$

$$OD^2 = 78.52$$

$$OD = \sqrt{78.52} = 8.86 \text{ cm}$$

$$CD = 8.86 - 6 = 2.86 \text{ cm}$$

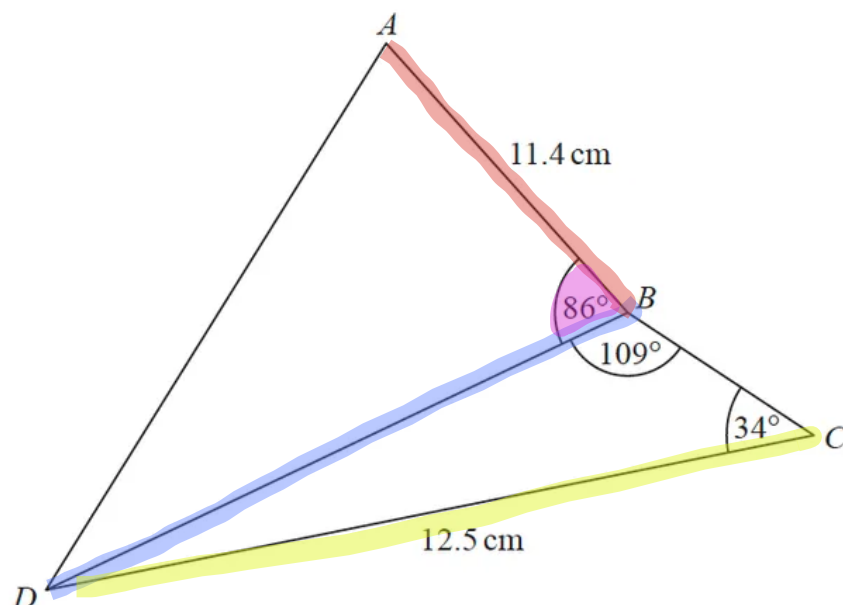
$$\text{Arc } ABC = \frac{220}{360} \times 12\pi = 23.04 \text{ cm}$$

$$\text{Perimeter} = 23.04 + 2.86 + 14 = 39.898 \text{ cm}$$

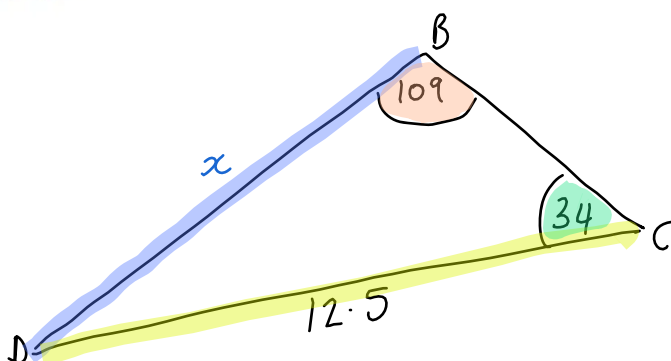
(Total for Question 16 is 5 marks)

[Edexcel, GCSE, Autumn 2018, Paper 3H, Q16]

5



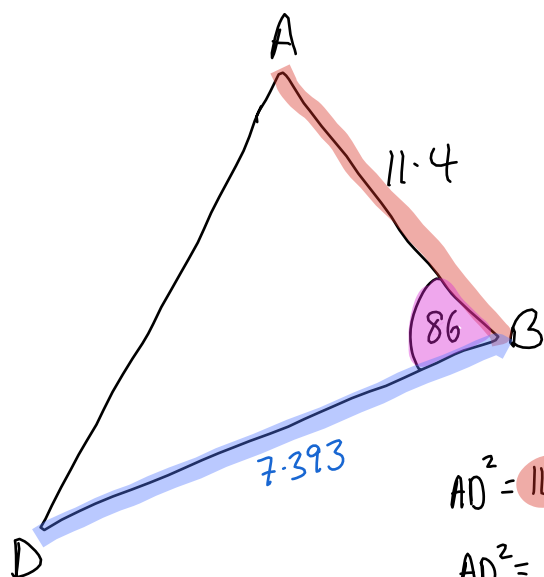
Work out the length of AD .
Give your answer correct to 3 significant figures.



$$\frac{x}{\sin(34)} = \frac{12.5}{\sin(109)}$$

$$x = \frac{12.5 \sin(34)}{\sin(109)}$$

$$x = 7.393$$



$$AD^2 = 11.4^2 + (7.393)^2 - 2 \times 11.4 \times 7.393 \times \cos(86)$$

$$AD^2 = 172.85$$

$$AD = \sqrt{172.85} = 13.147$$

..... 13.1 cm

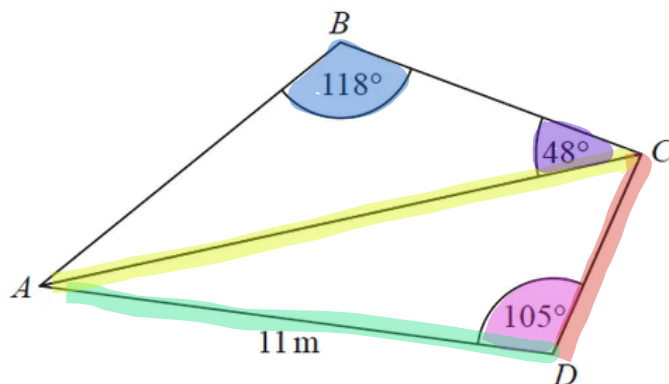
(Total for Question 17 is 5 marks)

[Edexcel, GCSE, Summer 2018, Paper 3H, Q17]

The Cosine Rule (Questions from Past Papers)

6

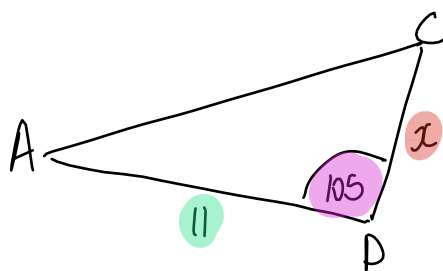
ABC and ADC are triangles.



The area of triangle ADC is 56 m^2

Work out the length of AB .

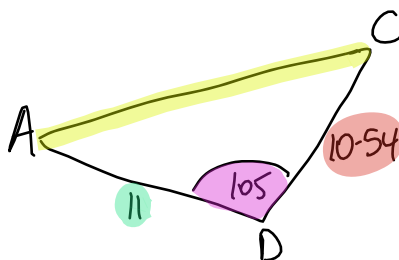
Give your answer correct to 1 decimal place.



$$\frac{1}{2} \times 11x \times \sin(105) = 56$$

$$x \times 11 \sin(105) = 112$$

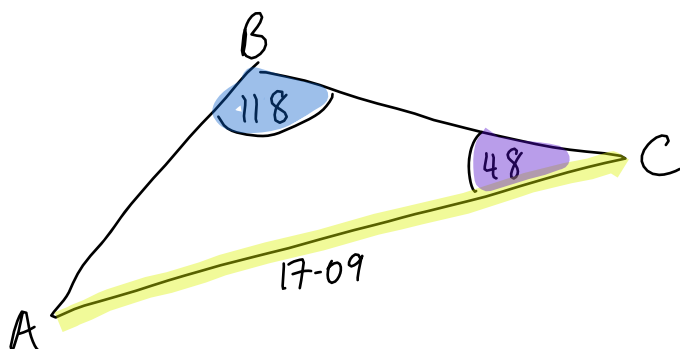
$$x = \frac{112}{11 \sin(105)} = 10.54$$



$$AC^2 = 11^2 + 10.54^2 - 2 \times 11 \times 10.54 \times \cos(105)$$

$$AC^2 = 292.106$$

$$AC = \sqrt{292.106} = 17.09$$



$$\frac{AB}{\sin(48)} = \frac{17.09}{\sin(118)}$$

$$AB = \frac{17.09 \sin(48)}{\sin(118)} = 14.384$$

14.4 m

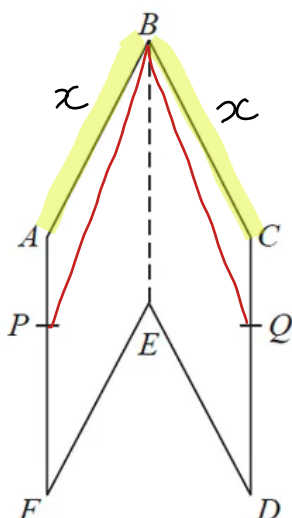
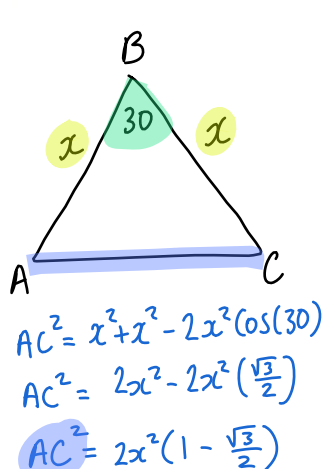
(Total for Question 17 is 5 marks)

[Edexcel, GCSE, Autumn 2017, Paper 3H, Q17]

The Cosine Rule (Questions from Past Papers)

7

The diagram shows a hexagon $ABCDEF$.



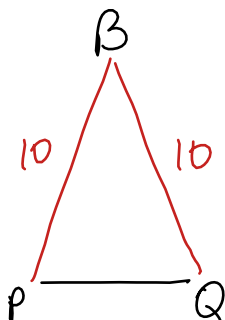
$$AC = PQ$$

$$AC^2 = PQ^2 = 2x^2 \left(1 - \frac{\sqrt{3}}{2}\right)$$

$ABEF$ and $CBED$ are congruent parallelograms where $AB = BC = x$ cm.
 P is the point on AF and Q is the point on CD such that $BP = BQ = 10$ cm.

Given that $\text{angle } ABC = 30^\circ$,

prove that $\cos PBQ = 1 - \frac{(2 - \sqrt{3})x^2}{200}$



$$PQ^2 = 10^2 + 10^2 - 2 \times 10 \times 10 \times \cos PBQ$$

$$2x^2 \left(1 - \frac{\sqrt{3}}{2}\right) = 200 - 200 \cos PBQ$$

$$-200 + 2x^2 \left(1 - \frac{\sqrt{3}}{2}\right) = -200 \cos PBQ$$

$$\frac{-200}{-200} + \frac{2x^2 - \sqrt{3}x^2}{-200} = \cos PBQ$$

$$1 - \frac{(2 - \sqrt{3})x^2}{200} = \cos PBQ$$

(Total for Question 22 is 5 marks)

[Edexcel, GCSE, Summer 2017, Paper 1H, Q2]